SYSDESIGN REFERENCE

Distributed ID Generation Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

Create unique, scalable identifiers with acceptable ordering, coordination, and information leakage.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / ID-GENERATION

SYSDESIGN

Distributed ID Generation Tradeoff Matrix

CORE IDEA

Create unique, scalable identifiers with acceptable ordering, coordination, and information leakage.

Reference architecture decision chain

01	02	03	04
Uuid Variants	Database Sequences	Snowflake-Style Ids	Clock Behavior

Decisions to defend

+ UUID variants
+ database sequences
+ Snowflake-style IDs
+ clock behavior
+ and shard locality.

Failure modes to test

+ Clock rollback
+ duplicate worker IDs
+ hotspotting
+ and guessable IDs.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Distributed ID Generation Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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